

Keara L. Hill

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EDUCATION

State University of New York at Binghamton
Doctorate of Philosophy in Computer Science
Masters of Science in Computer Science
Bachelor of Science in Computer Science

Binghamton, NY
expected May 2027
May 2022
May 2016

RESEARCH INTERESTS

Formal probabilistic methods for software security, vulnerability analysis, exploitability analysis

RESEARCH EXPERIENCE

Binghamton University
Graduate Assistant
Student Assistant

Binghamton, NY
August 2024–Present
May 2024–August 2024

- Investigated formal methods for analyzing timing side channel vulnerabilities
- Developed tool implementing abstract interpretation-based side channel analysis technique on LLVM IR
- Applied tool to real-world cryptographic algorithms to quantify timing leakage
- Supported other students on systems security projects which used introspective hypervisor technology

TEACHING EXPERIENCE

Binghamton University
Teaching Assistant

Binghamton, NY
August 2024–May 2024

- Helped new professors design curricula for graduate-level Programming Languages and Compiler Design courses
- Developed automatic grading software that built and tested student projects written in C, Java, and Haskell
- Provided grading assistance in graduate level Programming Languages, Compiler Design, Design Patterns, and Information Security courses
- Supported students that struggled with material or projects

PROFESSIONAL EXPERIENCE

Grammatech
Software Engineer

Ithaca, NY
October 2019–February 2023

- Developed feature selection and results analysis techniques for a machine learning-based analysis of binary rewriting tools
- Architected a production-quality refactor of a prototype embedded library matcher
- Responsible for maintenance, improvements, and testing for the primary component of the CodeSentry library matching backend
- Contributed to productization of CodeSentry library detector from its original research project
- Continuously evaluated and improved on a suite of string-based library detectors for COTS binaries

Assured Information Security
Software Engineer

Rome, NY
September 2015–September 2019

- Led a multi-institution research team through design and development of a prototype assured hardware compiler
- Reviewed literature to determine the state of the art for assured hardware generation
- Designed a language for dynamic function call pattern matching
- Developed custom software suite for cyber warfare exercises
- Mentored a research team investigating formal software verification for mission-critical systems
- Supported a full-scale cyber warfare exercise with over twenty participants
- Contributed to development lifecycle of large, multi-language projects

- Ensured software compatibility with a variety of current and legacy systems

Co-op

May 2015–August 2015

- Reviewed literature to support research into cyber deception techniques
- Developed prototype cyber deception tools using the IntroVirt introspective virtual machine

PUBLICATIONS

The 7th Annual Symposium on Machine Programming (MAPS)

December 2023

Predicting the Success of x86-64 Binary Rewriters

- Used machine learning techniques to predict whether a binary was transformable for a suite of binary rewriting tools
- Discovered, investigated, and extracted features likely to predict transformation success

Master's Thesis

Spring 2022

A Formal Role-Based Access Control Model for a Microservice-Based Malware Analysis Platform

- Designed access control model for platform with known malicious and untrustworthy actors
- Proved security properties of model

PRESENTATIONS

Upstate Programming Languages Seminar

August 2025

Probabilistic Abstract Interpretation for Quantifying Side Channel Leakage

Cornell University

- Presented ongoing side channel analysis research and preliminary results

SERVICE

OOPSLA

2026

Member, Artifact Evaluation Committee

OOPSLA

2025

Member, Artifact Evaluation Committee

TECHNICAL SKILLS

Programming Languages: C, C++, Go, Python, C#, PowerShell, Haskell, x86 assembly, Prolog, AWK, bash, Java, SML, BLISS

Operating Systems: Linux, Solaris, OpenBSD, Microsoft Windows, OpenVMS, RSTS/e

Tools: Frama-C, Intel Pin, Antlr 4, Wireshark, Cobalt Strike, HOL4, $\LaTeX$, MongoDB